Andrew Thompson

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San Diego, California | U.S. Citizen

EDUCATION

University of California, San Diego *Expected Graduation June, 2027*
B.S. Mechanical Engineering with specialization in Renewable Energy **3.8/4.0 GPA**
- Summer 2025: Study Abroad @ Yonsei University in Seoul, South Korea

EXPERIENCE

Varda Space Industries
Vehicle Systems Integration & Test Engineering Intern *Sept. 2026 - Dec. 2026*

- Authored specialized N_2 fill, vent, and **leak-check** procedures and **test setups** for orbital satellite bus bi-propellant diaphragm tanks, **mitigating flight hardware risk**, enabling technician execution
- Closed company test-safety gap by developing **Excel** stored energy calculator (pressure/volume to TNT equivalent per NASA and OSHA standards) to establish safe blast radius for all personnel
- Supported full satellite bus **V&V** campaigns (**vibration, mass properties, and sensor checkouts**)

Students for the Exploration and Development of Space (SEDS@UCSD)

Director of Internal Affairs *May 2026 - Present*

- Served as a member of the executive board, directing recruitment, evaluating **300+ applicants** to select, onboard, and mentor 30 new members across prop, structures, avionics, and controls subteams

Lander Lead Fluids Engineer *Jan. 2026 - Present*

- Managed **7 engineers** in design/manufacturing of propellant feed systems of **world's first** collegiate pressure fed **methalox**, 2,000 lbf thrust, 16ft tall, **self-landing** rocket, dedicating **25+ hrs/week**
- Mentored new members on Lander's fluid systems, enabling them to take **immediate ownership** of component design and manufacturing, facilitating **knowledge transfer**
- Synchronized flight-critical **solenoid** actuation timing and **PT/TC** sensor acquisition with Avionics & Controls via **standardized documentation** and cross-team **communication**
- Coordinated **200psi** vehicle **leak check** with custom test setup, and led following **troubleshooting**
- Optimized component sourcing (valves, **AN, NPT, Swagelok fittings**) organizing full system **BOM**

Lander Fluids Engineer *June 2025 - Jan. 2026*

- Assembled full propellant feed system, prioritizing **reusability** and **affordability**, accounting for extreme **thermal conditions**, **collapse factor**, **boil off**, and pressure budget
- Authored and maintained custom **P&ID** & **Fitting Diagram** with over **30 unique components**
- Developed a custom **MATLAB** script to calculate system head loss, accounting for **major/minor losses**, ensuring propellant **pressures** and **mass flow rates** meet engine requirements

Colossus Fluids Engineer *Jan. 2025 - June 2025*

- Restored a **3,000 lbf bi-propellant static fire test stand** after FOD failure by redesigning high-pressure fluid lines (stainless steel **tube cutting, bending, flaring, assembly**)
- Manufactured custom components with **SolidWorks**, **3D printing**, and **CNC/manual** machining for structures and electronics integration
- Engineered and plumbed an E-STOP triggered **N_2 based fire suppression system**

SKILLS

Software: P&ID (LucidChart), MATLAB, Excel, Microsoft Office, CAD (familiar), Jira, Confluence
Manufacturing: Tube assembly, Swagelok, AN, NPT, Soldering
Soft Skills: Personable, leadership, communication, teamwork, collaboration, organized, optimistic